

Multi-resolution semantic abstraction over the TKH hypergraph

Ilia Dzneldze, 2026-09-23. I designed and directed this project and used a coding agent (Claude Code) to implement, run and debug it; `AI_USAGE.md` has the specifics. The reasoning behind each choice, and every number below in more detail, is in `DESIGN_NOTES.md` (cited as DN §n).

1. The data and its snapshots (T1)

I cut the export at the hyperedge `year`: a snapshot at `t` keeps every edge dated `t` or earlier and the nodes those edges reference.

snapshot	edges	nodes	concepts	papers	new nodes	arity > 2	seen after t
≤2020	405	1,527	1,406	16		73%	22
≤2022	589	2,200	2,046	23	673	76%	35
≤2024	1,063	4,184	3,895	39	1,984	80%	19
≤2026	1,429	5,798	5,428	52	1,614	80%	0

Median arity is 3 throughout and the maximum grows from 45 to 65. The node count grows 44%, then 90%, then 39%, so the corpus nearly doubles between 2022 and 2024. Three data-quality problems matter later. `origin_year` is missing for 86% of nodes, so it can't drive anything. There are 307 duplicate surface forms among concept nodes, because there is no entity resolution: "MAE" is a separate node in each paper that mentions it. The last column counts nodes an early edge references before the corpus has "seen" them (`first_seen_year` after `t`). Those break the temporal-honesty guarantee I first claimed (§5.3). Articles and authors stay in the snapshot as edge members but are never clustered (DN §1).

2. Formal statement (Deliverable 0)

Construction. For snapshot `t` let V_c be its concept nodes; for a hyperedge `e` let e_c be its concept members and $n_e = |e_c|$; let x_i be the unit MPNet embedding of node `i`'s surface form. For $i \neq j$:

```
S(i,j) = sum over hyperedges e with i, j in e_c of 1/(n_e - 1)
M(i,j) = cos(x_i, x_j) if j is in kNN_15(i) or i is in kNN_15(j), else 0
A(i,j) = alpha * S'(i,j) + (1 - alpha) * M'(i,j),    alpha = 0.3
d(C,D) = 1 - mean of A(i,j) over i in C, j in D      (missing entry = 0)
```

S' and M' are each graph divided by its own 99th percentile and clipped to $[0,1]$. S is a weighted clique expansion: every member spreads one unit of weight over its co-members, so an edge of arity n adds $n/2$ in total rather than $C(n,2)$. Average linkage merges the pair with the smallest d , and level k is the cut of the dendrogram with at most N_k groups, $(N_0, N_1, N_2) = (12, 50, 200)$.

Criterion, not objective. Average linkage is a greedy merge rule and optimises nothing globally. What it satisfies is a criterion: at every step, merge the two groups with the highest mean pairwise affinity, and level k is the state of that process at N_k groups. What the criterion buys is nesting. With `missing = 0` the merge heights never decrease, so the cuts are laminar by construction. The implementation matches `scipy`'s dense average linkage on heights and cuts, and the test fails for the weighted variant, so it isn't trivial (DN §8).

Trade-off (P3). One joint graph makes the structure/meaning balance a single sweepable number, and a strong semantic signal can override weak structure at any level. I rejected a two-stage scheme (structure fixes the coarse level, semantics refines it) because two topically identical but citation-disconnected groups could then never merge at the top. I did not implement it to compare, and that is a gap. The thing I got wrong is what α measures. The two normalised graphs sit on different scales (median entry 0.06 for S' , 0.57 for M'), so at $\alpha = 0.3$ structure carries **6.6%** of the affinity mass, not 30% (DN §17). Structure still matters more than that number suggests, because 94% of structural pairs are pairs the k -NN graph never proposes. But the honest description is semantic k -NN clustering with the hypergraph as a tie-breaker.

Complexity. Building S is $O(\text{sum of } n_e \text{ squared})$, 66k pairs at 2026. M is a $k=15$ neighbour search, and the heap-based merge is about $O(E \log E)$ while neighbour lists stay sparse. The four-snapshot pipeline takes about

90 s on a laptop CPU, and nearly all of that is the embedding pass.

property	status	evidence
P1 laminar	by construction	exact check at every snapshot (<code>validate_hierarchy.py</code>)
P2 size budget	by construction	at most N_k groups per cut; $N_0 = 12$ everywhere
P3 coherence	empirical	blind intruder test passes at levels 1 and 2, not at level 0 (§5.1)
P4 hyperedge fidelity	partial	S is a projection; T4 collapse written, not used to build levels (§4)
P5 stability	empirical, not enforced	ARI 0.35/0.45/0.65 by level; change not localised (§5.2)
P6 faithful labels	empirical	0 of 48 rated wrong blind; input temporally honest for ~98% of nodes (§5.3)

3. Method in the literature

The weighting follows the reduction of hypergraph Laplacians to clique and star expansions (Zhou, Huang & Schölkopf 2006; Agarwal, Branson & Belongie 2006). I borrow the arity-aware weighting and skip the spectral step. For the hierarchy I chose plain agglomeration over principled hierarchical models. Clauset, Moore & Newman (2008) fit a dendrogram as a generative model and score it on held-out links, the same logic as my held-out-hyperedge test (§5.1). Peixoto's nested block model (2014) picks the number of groups per level from the data, which would remove the fixed cut sizes I suspect cause global churn (§5.2) but give up the hard budget P2. I rejected hypergraph modularity (Kumar et al. 2020) because it does not produce a laminar family without extra repair. For time, the event vocabulary is from Greene, Doyle & Cunningham (2010), and my matching scheme is what Asgari, Cazabet & Borgnat (2023) call No-Smoothing: cluster each snapshot, then match by Jaccard. They found it the least smooth option in most settings; my warm-start test (§5.2) found the smoother option costs some coherence. Ruggeri, Lonardi & De Bacco (2023) show hypergraph communities are hard to detect when pairs rarely recur across edges; here only 1.7% of concept pairs recur across papers, which I think is why structure generalises weakly. Loukas (2019) coarsens pairwise graphs, so never has to decide what an edge spanning three or more groups becomes, the part T4 decides. SHyPar (Sajadinia et al. 2024) coarsens hypergraphs spectrally and argues local heuristics like mine lose global structure, a plausible but untested reading of why my multilevel variant collapsed (§4).

4. Temporal coupling (T3) and hyperedge collapse (T4)

T3. Each snapshot is clustered independently and matched to the next by Jaccard overlap on shared nodes. Persistent ids carry forward and 1,091 events (birth, death, stable, grow, shrink, merge, split) go to `temporal_events.json`. Matching keeps each snapshot reproducible on its own, at the cost of never pushing the clustering toward stability. Of the three thresholds, only `MATCH_THRESHOLD` is sensitive. Moving it from 0.10 to 0.20 changes deaths from 12 to 78 and splits from 192 to 84 (DN §10).

T4. An edge whose endpoints all fall in one super-node becomes internal and is kept as a per-super-node count. One spanning two super-nodes becomes a pairwise edge. One spanning three or more stays a single k-ary edge rather than being exploded into $C(k,2)$ pairs, which would inflate density and turn one asserted fact into many. Coarse edges on the same super-node set merge into one weighted edge with a relation-type breakdown, which is what the rule loses: which original member was in which original edge. At level 0 in 2026 the 1,429 edges become 592 coarse edges (128 pairwise, 464 k-ary) and only 7 internal, few because articles pass through as singletons. The rule runs at every level and is written to `hierarchy.json`, but the shipped levels are cuts of one dendrogram, so nothing reads it back, which falls short of what the brief asks. A variant that clusters each coarser level on the collapsed hypergraph below it lost a third or more of level-0 coherence and put about 40% of nodes into one cluster, so under the rule I had set beforehand it did not ship (DN §15).

5. Evaluation

5.1 Coherence and the circularity hazard

Measuring coherence with the MPNet embeddings that drive clustering would grade the method with its own ruler, so I measured it three other ways. The first is TF-IDF over the same text, against a random-labels null. The observed values are far from chance ($z = 709, 864$ and $1,424$ by level), but this is the weakest check. TF-IDF reads the same strings, and 69% of MPNet neighbour pairs share a TF-IDF term against 7% of random pairs (DN §5), so it is a second view of the same signal. A degree- and arity-preserving shuffle of the hypergraph gets almost the same TF-IDF coherence ($z = 0.3, 0.4$ and 2.9), which fits structure carrying 6.6% of the mass.

The second check doesn't read text at all: hide 20% of hyperedges, cluster on the rest, and ask how often members of a hidden edge share a cluster, relative to a size-matched random partition. At level 0, moving alpha from 0 to 0.3 to 0.5 raises this lift from 1.95 to 2.42 to 3.53. Holding out whole papers cuts that to 1.98, 2.11 and 2.54, so structure mostly encodes co-occurrence within papers it has already seen. It clears the bar I set only at level 0 (paired gain CI 0.01 to 0.25), and no alpha beats 0.3 on both axes (figure 1).

Structure vs. meaning across alpha (2026, 5 seeds, 20% held out)

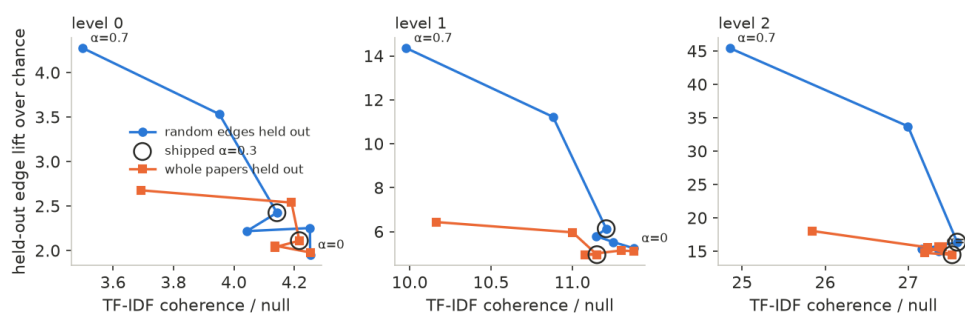


Figure 1. The structure/meaning trade-off across alpha: held-out hyperedge lift against TF-IDF coherence over its null, holding out random edges (blue) or whole papers (orange). The ring marks alpha = 0.3.

The third check is a blind intruder test in the style of Chang et al. (2009). A fresh LLM rater with no repo access, and a different model from the labeller, sees five members of a super-node plus one type-matched node from another level-0 branch, and picks the odd one out. Chance is 1 in 6. On null items made from random nodes it found the intruder 1 time in 20, so it was not reading surface cues. On real items it found it 17 of 23 times at level 2 and 14 of 24 at level 1, both well above chance, but only 4 of 12 at level 0 (CI 14% to 61%). That fails the rule I set beforehand (figure 2). My conclusion is that levels 1 and 2 hold together for a blind reader, and level 0, which is also the least stable level, is not shown to.

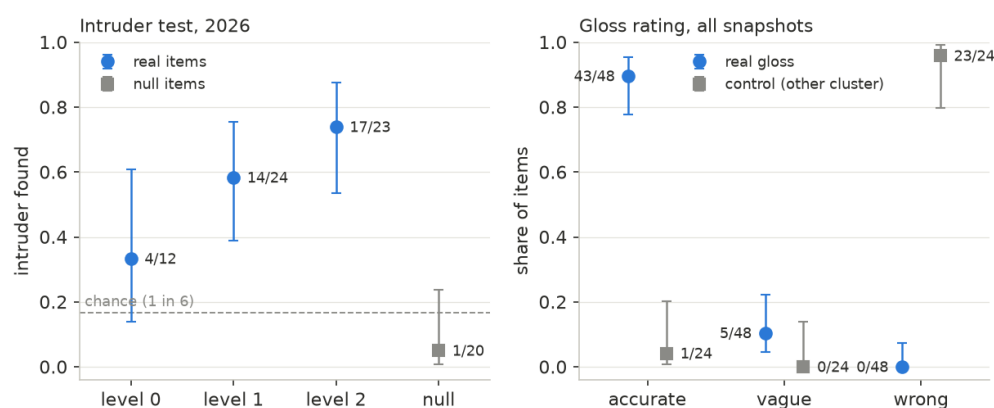


Figure 2. Left: intruder detection by level with Wilson 95% CIs, against chance and against null items. Right: blind ratings of glosses shown with their own cluster's unseen members (blue) or another cluster's (gray).

5.2 Stability

Rebuilding five times with 10% of hyperedges removed gives ARI 0.58, 0.77 and 0.91 by level against the original. This only ever disturbs the structural graph, so it rewards ignoring the hypergraph (0.75, 0.92 and 0.98 at alpha = 0.05), and I don't read it as robustness. Cross-snapshot ARI on shared nodes is 0.35, 0.45 and 0.65, with 95% CIs from resampling half the shared nodes (figure 3). (My first version resampled with replacement, which inflates ARI for small clusters; DN §20.) Per transition, the intervals at level 1 don't overlap: 0.52, then 0.37 over 2022 to 2024, when the corpus nearly doubled, then 0.47. So transitions differ for real, and the three-transition mean is a summary. Change is not localised where new hyperedges land: at no level does churn rise with a

cluster's share of new edges (Spearman -0.32 , $+0.03$, -0.01 , CIs spanning 0), and 351 fine clusters with under 2% new edges still churn 0.44. It does follow new semantic neighbours at levels 1 and 2 (Spearman 0.42, 0.18), but I found that only after the planned test failed, so it is a lead. A warm-started variant raised level-2 cross-snapshot ARI from 0.65 to about 0.9, at a coherence cost that is real but noisy (level-2 z-scores 7% to 18% lower on average), so by the rule I had set it didn't ship, though it came closer than I first wrote.

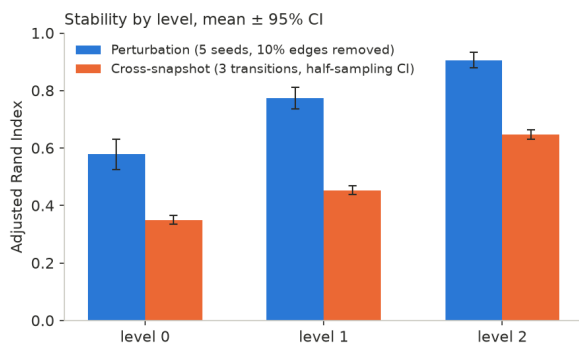


Figure 3. Stability by level: perturbation (5 seeds, t-interval) and cross-snapshot (half-sampling interval over shared nodes).

5.3 Label faithfulness

The 248 labels and glosses were written by fresh LLM agents from a seeded sample of 25 members plus type counts (prompts in `labeling_input.json`); an earlier set, written by agents that could see the question files, was replaced (DN §12). The NLI check grades each gloss against up to 15 members the labeller never saw, with a control that uses another cluster's members. Contradiction runs 5% to 14% by snapshot (controls 45% to 63%), and "not entailed" runs 60% to 79%. I didn't know which number was the over-claim rate, so I had a blind rater mark 72 items as accurate, vague or wrong, with real and control items mixed (figure 2, right). It rated **0 of 48 real glosses wrong (CI 0% to 7%)** and 23 of 24 controls wrong. It also called 32 of the 36 real glosses that NLI marks "not entailed" accurate. By the rule I had set, the not-entailed rate measures a weak premise, not over-claim. NLI contradiction agrees with the rater's "wrong" only moderately ($\kappa = 0.46$, CI 0.24 to 0.68) and raises false alarms. Grading glosses against their members' source-paper titles, which neither labeller nor clustering saw, points the right way (21% contradicted against 31% for controls) with overlapping intervals, so I don't rely on it. Temporal honesty holds for the labeller's input for about 98% of nodes, not by construction. The 2020 label `L1_S00048` names EquiformerV2, which the corpus first saw in 2024.

5.4 Extrinsic: coarse-to-fine retrieval

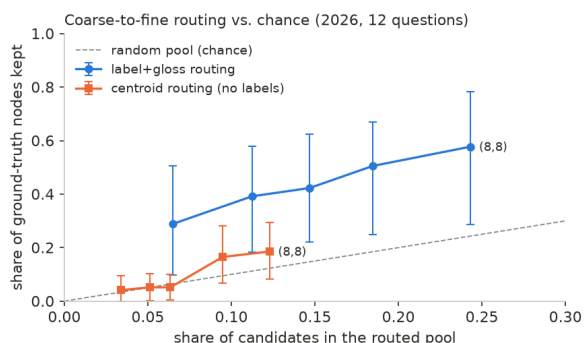


Figure 4. Share of each question's ground-truth nodes kept in the routed pool against the pool's share of candidates. Label routing (blue), centroid routing (orange), random pool (dashed); bootstrap CIs over 12 questions.

For each type-A question, a drill-down routes through label and gloss text (top 8 of 12 level-0 groups, then top 8 of their level-1 children) and ranks the nodes in the resulting pool; the flat baseline ranks all 3,104 method-like nodes by cosine to the question. Ground-truth names resolve to nodes by exact or substring match. The first matcher let element nodes ("N", "S", "Si") match names like "physics-informed", so a third of the answer key was chemical elements. With both sides of a substring match required to be at least 4 characters, 12 questions keep a ground truth (DN §16). Routing works: the pool holds 58% of ground-truth nodes while covering 24% of candidates, a lift of 0.33 over a random pool of the same size (CI 0.04 to 0.54), and every budget clears chance. Routing on member centroids instead of labels does not (figure 4). Recall@20 does not improve. Drill-down gets

0.121 against flat's 0.119 in-sample, and with settings chosen by leave-one-out the paired difference is -8.7 points (CI -25.9 to $+0.7$, $p = 0.37$). The whole comparison rests on 4 versus 5 retrieved ground-truth nodes. The hierarchy finds the right region, and the ranking inside it is the weak part.

5.5 What I would ship

The single dendrogram at $\alpha = 0.3$, with matching-based tracking. The multilevel T4 variant and the warm start each bought stability with coherence, clearly for the first and noisily for the second, so the warm start is the one to revisit. α was picked on the coherence and stability numbers above, so those are in-sample. Without the confounded perturbation measure, 0.3 is still the only value beating 0.5 on every per-level number, though five others between 0.05 and 0.45 beat it on five of six.

6. Verified and assumed

Verified, each with a check or a number behind it: laminarity at every snapshot; the UPGMA implementation against scipy; coherence against two nulls, held-out hyperedges and a blind intruder test; stability CIs by half-sampling, with the old estimator's bias reproduced in a test; the NLI judge checked against blind gloss ratings; the structural share of the affinity (6.6%) measured rather than assumed; the ground-truth matcher fixed and pinned by a test; a one-command rerun from a fresh environment that reproduces every output byte for byte.

Assumed or unresolved: the two-stage alternative is argued, not tested; blind judgements come from one LLM rater, with no second rater or human pass; why α near 0.3 works, rather than only that it does; the temporal thresholds, picked by feel, with `MATCH_THRESHOLD` sensitive; whether fixed cut sizes cause the global churn; the 7 substring matches in the answer key, unchecked against the papers; the Linux install, not run end to end.

Reproducing it. `py -3.11 scripts/reproduce_all.py` goes from a fresh clone to every result and figure in one command, building its own environment from the pinned requirements first. From a folder with no environment at all, it reproduced all 45 output files byte for byte against the published checksums in 72 minutes on a laptop CPU. My model outputs are published too, so a different result elsewhere can be traced to the embeddings or the code (DN §23), and a reproducer can swap in their own labels and ratings (DN §24).

7. With four more weeks

Method: normalise the two graphs so α means influence and re-sweep, and cut at a fixed merge height rather than fixed counts, which I suspect drives the global churn. Evaluation: a human rater for the intruder and gloss tasks, a semantic-side perturbation so stability stresses both signals, and the two-stage alternative measured rather than argued. Extrinsic: map type-B questions and method claims to nodes so there are more than 12 questions, then a better ranker inside the routed pool.

References

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